Task 08: Evaluate CPU States and Protection

Operating Systems (CSA04) — Assignment 1 (CO1)

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Marks: 3 Marks (Bloom's: K5 - Evaluate)
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> ***Task 8 Description:*** *Evaluate User mode vs. Kernel mode, privileged instructions, and the hardware mode bit. Explain interrupts, traps, and timer ticks. Discuss the consequences of allowing arbitrary programs to disable interrupts. Contrast interrupt-driven processing with polling.*

1. DUAL-MODE HARDWARE OPERATION & PRIVILEGE LEVELS

CPUs enforce privilege boundaries using hardware mode bits (e.g., CPL bits in the CS register on x86, or CurrentEL on ARMv8):

- **User Mode (Ring 3):** Application code runs with restricted CPU capabilities. Direct access to hardware I/O ports, page table modifications, and execution of privileged instructions (e.g., `cli`, `hlt`, `wrmsr`) are prohibited. Attempting to run a privileged instruction triggers a General Protection Fault (trap to kernel).
- **Kernel Mode (Ring 0):** The operating system has unrestricted access to hardware instructions, control registers, and physical memory.
- **Timer Interrupt:** A hardware timer (APIC or ARM generic timer) generates periodic interrupts at set intervals (e.g., every 1ms). This switches the CPU to Kernel Mode and invokes the scheduler, ensuring runaway processes cannot monopolize the CPU and preventing missed alarm deadlines.

2. CONSEQUENCES OF DISABLING INTERRUPTS ON BEDSIDE MONITORS

If user programs could execute privileged instructions like `cli` (Clear Interrupt Flag):

A misbehaving UI component or third-party telemetry driver could disable interrupts during a long calculation or deadlock. Because hardware interrupts from ECG and SpO2 sensors are masked, the CPU will drop incoming sensor data. The 10ms alarm deadline would be missed, potentially failing to notify clinical staff of a critical cardiac event.

3. HARDWARE MEMORY PROTECTION: MMU & PAGING

The hardware Memory Management Unit (MMU) uses page tables with protection flags (User/Supervisor, Read/Write, No-Execute) to validate memory access. Loading a process base address into `CR3` establishes address space isolation. Any attempt to read or modify another process's memory triggers a Page Fault exception (vector 14), allowing the kernel to terminate the offending process with `SIGSEGV` while keeping other systems operational.

4. ALTERNATIVE CONCEPT: INTERRUPT-DRIVEN VS. POLLING SENSOR ARCHITECTURE

Evaluation Metric	Interrupt-Driven Processing (Adopted)	Continuous Sensor Polling (Alternative)
Worst-Case Alarm Latency	**< 50µs:** Hardware asserts an IRQ line, triggering immediate ISR execution.	**Unbounded or Dependent on Interval:** $T_{latency} = T_{poll_interval} + T_{process}$. Delays can reach 10–20ms.
CPU Utilization & Heat	**Zero idle load (< 2%):** CPU stays in low-power sleep (`HLT` / `WFI`) until an event occurs.	**100% Core Load:** Busy-waiting drains tablet batteries and generates excess heat.